

Attention-Enhanced U-Net for Automated Image Colorization

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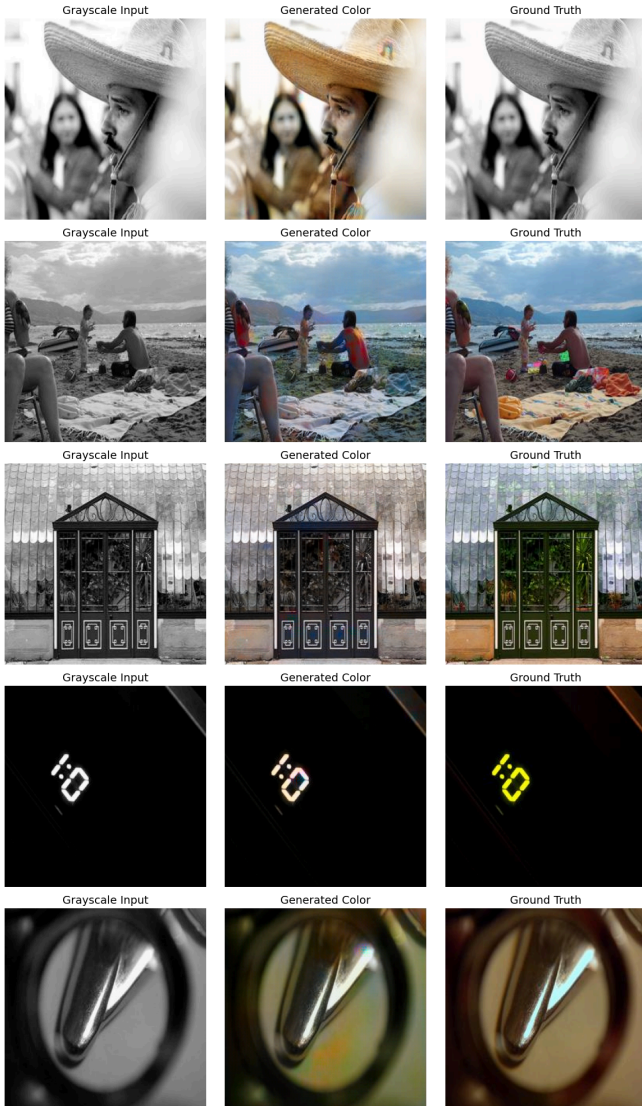


Figure 1. Colorised Images resulting from Version 1.6 of the Implementation

1. Task

The primary objective of this project is to perform fully automated image colourisation, transforming monochrome grayscale inputs into visually realistic, vibrant, and structurally coherent colour versions. The central difficulty of this task lies in the multi-modal ambiguity of colours; an object such as the sky can be legitimately colored in various shades of blue, grey, or sunset-toned gradients. Traditional machine learning models frequently struggle with this multi-modal ambiguity, often mathematically averaging the

possibilities to minimize absolute pixel error, which inadvertently results in blurry, desaturated, or sepia-toned images.

Our core objective was to build a generative system that prioritizes adversarial stability, successfully prevents ‘colour bleeding’ across complex textural boundaries, and critically balances rigid structural metrics (like SSIM) against human-perceived chromatic realism. To achieve this, the project required systematically identifying and addressing normalization bottlenecks, refining colour-space mapping techniques, and implementing dynamic learning rate adjustments to prevent mode collapse during adversarial training.

2. Related Work

Image colourisation is an active research area that has been largely automated through algorithms designed to assign colors based on learned spatial patterns and textures. Recent advancements have seen the development of a wide variety of approaches, including diffusion models, transformer architectures, convolutional neural networks (CNNs), and Generative Adversarial Networks (GANs). As we iterated on our architecture to solve training instability and color-space mapping failures, we integrated insights from the following related works:

- **GAN Stabilization via TTUR:** We incorporated the Two Time-Scale Update Rule (TTUR) proposed by Heusel et al. (2017), which demonstrates that GANs converge to a local Nash equilibrium more reliably when the discriminator learns at a slower, distinct learning rate compared to the generator.
- **Group Normalization:** Because our generator relies on a deep ResNet backbone, we reviewed Wu & He’s (2018) work on Group Normalization, discovering it preserves global color statistics without the batch-to-batch contamination seen in standard Batch Normalization.
- **Spectral Normalization:** To strictly enforce the Lipschitz constraint on our discriminator and prevent gradients from exploding or vanishing, we adapted the Spectral Normalization techniques introduced by Miyato et al. (2018).

3. Approach

To optimize computational efficiency and model stability, we process images using the CIELAB (LAB) color space rather than the standard RGB format. The L channel represents lightness (the grayscale input), while the a and b channels represent color values. The L channel is normalized to a range of $[-1, 1]$, and the ab channels are scaled by a factor of 110.0 to simplify the base computation.

Generator Architecture

Our generator is an Attention-Enhanced U-Net, functioning as a Convolutional Encoder-Decoder.

Encoder: Initially built from scratch with 5 convolutional blocks, we upgraded the encoder to a pre-trained ResNet34 backbone utilizing IMAGENET1K_V1 weights. This transfer learning approach completely replaces the custom encoder, allowing the network to extract highly robust spatial features without wasting critical epochs attempting to learn basic edge detection.

Decoder: The decoder progressively upsamples the image representations. To resolve spatial mismatches caused by the ResNet34 stride mechanics, an additional final upsampling block was engineered to safely restore the image to the required 256×256 resolution without losing structural fidelity. We integrated Custom Attention Gates on the skip connections. Instead of blindly copying data directly from the encoder, these attention gates actively filter the feature maps, allowing the network to heavily focus on salient foreground objects while actively suppressing irrelevant background noise.

Discriminator Architecture

Instead of evaluating the entire generated image as a single global "Real/Fake" value, we utilize a PatchGAN discriminator. This network specifically evaluates local $N \times N$ patches, heavily penalizing the generator for producing unrealistic local textures and successfully enforcing crisp, high-frequency details. We wrapped the discriminator's convolutional layers in PyTorch's `spectral_norm` to mathematically enforce a gradient speed limit and prevent GAN mode collapse. Additionally, all normalizations were transitioned to Group Normalization (8 groups) to isolate math per image and prevent batch-induced color smearing.

Dynamic Loss Functions and Optimization

Our objective function underwent iterations to solve the problem of "Safe Color Collapse." Standard L1 loss calculates strict pixel-by-pixel mathematical differences, which often leads to models generating muddy or blurry colors to minimize absolute penalties. We initially implemented a VGG16 perceptual loss network to compare high-level feature maps. However, because VGG16 is trained exclusively on RGB images, feeding it LAB tensors caused a massive color-space mismatch that actively punished the generator for producing true, vibrant colors.

We completely removed the VGG network and shifted to a rebalanced GAN formulation. To force the model to take risks rather than defaulting to mathematical grays, we initially halved the strict L1 weight to 50.0. For our final optimization strategy, we transitioned to a Smooth L1 (Huber) loss to safely fine-tune vibrant colors without facing destructive gradient spikes, and heavily utilized Discriminator Feature Matching to naturally regain structural crispness natively within the LAB color space.

Our generator is ultimately trained on the following objective function:

$$Loss_G = Loss_{GAN} + (\lambda_1 \times Loss_{GAN}) + (\lambda_2 \times Loss_{FM})$$

where $Loss_{GAN}$ drives overall adversarial realism, $Loss_{GAN}$ ensures global color accuracy without punishing vibrancy, and $Loss_{FM}$ enforces high-frequency textural details.

4. Datasets

We built a highly optimized dataset pipeline using a curated subset of the ImageNet-1K dataset. The dataset strictly consists of 10,000 images for the training set and 1,000 images for the validation and test sets.

To eliminate network latency and completely bypass data-loading bottlenecks during training, all images were heavily pre-processed. Every image was resized to exactly 256×256 pixels and immediately compressed into a single ZIP archive, which was deployed for high-speed local I/O extraction within our Colab runtime environments.

During the dynamic data-loading phase, these pre-resized images are converted natively from RGB into the LAB color space. To artificially expand the dataset and vastly improve the model's robustness against spatial transformations, we implemented data augmentation pipelines utilizing `torchvision.transforms`, specifically executing random horizontal flips and random rotations up to 15° .

5. Evaluation Metrics

We implemented a robust evaluation framework relying on three distinct metrics to capture both mathematical accuracy and human perception:

- **SSIM (Structural Similarity Index):** This metric specifically measures the perceived structural similarity between the generated images and the ground truth, calculating values based on luminance, contrast, and structure.
- **PSNR (Peak Signal-to-Noise Ratio):** This metric strictly quantifies the mathematical quality of the image reconstruction. Higher PSNR values indicate significantly lower pixel-level noise.
- **LPIPS (Learned Perceptual Patch Similarity):** Implemented in our final architecture, LPIPS evaluates human-aligned visual realism. Because SSIM and PSNR inherently reward muddy, average guesses, LPIPS utilizes deep network activations to closely mimic human visual judgment. A lower LPIPS score indicates higher perceived realism.

6. Results

We conducted rigorous empirical testing, iterating through six major versions of our model architecture to isolate and eliminate training instability, color washing, and artifact generation:

- **Version 1.0 to 1.1:** Our initial custom Attention U-Net (V1.0) suffered from high training overhead and semantic gaps in complex textural features, yielding an Average SSIM of 0.8965 and an Average PSNR of 20.8472. By upgrading to a ResNet34 Backbone and integrating a learning rate scheduler in V1.1, we effectively solved edge-detection waste. This immediately raised the SSIM to 0.9027 and PSNR to 22.2527.
- **Version 1.2:** In order to force better performance, V1.2 upgraded the encoder to ResNet50 and injected Self-Attention at the bottleneck layer to capture global context. This resulted in an immediate structural breakdown. The PSNR severely dropped by 10.5% down to 19.9087, and SSIM fell to 0.8789. The global context mixing caused aggressive "colour bleeding" (e.g., coloring the sky green to match grass) because standard Batch Normalization caused features to bleed across different images inside the batch.
- **Version 1.3:** To cure the bleeding, V1.3 reverted to the ResNet34 backbone and swapped to InstanceNorm2d. While this successfully isolated

the math to single images and stopped the color bleeding, it mathematically erased dominant background colors, leading to washed-out, grayish imagery. Quantitatively, the model rebounded to an SSIM of 0.9183 and PSNR of 23.4677.

- **Version 1.4:** We replaced Instance Norm with Group Normalization (8 groups), which successfully cured bleeding while preserving vibrant backgrounds. We also added TTUR to permanently handicap the Discriminator's learning rate, ensuring the Generator maintained a consistent advantage. V1.4 achieved our highest quantitative metrics with an SSIM of 0.9280 and PSNR of 23.8064. However, visually, it was a failure. The model suffered a "Safe Colour Collapse." Paralyzed by the massive L1 penalty ($\lambda = 100$), the network defaulted to generating muddy browns and sepia tones—the mathematical center of the color space—to minimize absolute risk.
- **Version 1.5:** To force the model to take risks and generate true color, V1.5 completely removed the mismatched RGB-trained VGG16 Perceptual Loss and halved the L1 weight to 50.0. This was a massive perceptual breakthrough. The model began confidently generating deep blue skies and accurate skin tones, deliberately accepting a minor drop in SSIM (0.9124) for massive gains in human-perceived chromatic quality.
- **Version 1.6:** In our final version, we regained structural crispness by implementing Discriminator Feature Matching natively inside the LAB space, forcing the generator to match intermediate discriminator layers rather than relying on external VGG networks. We reverted to Smooth L1 (Huber) Loss to allow the network to safely fine-tune vibrant colors without facing destructive gradient spikes. V1.6 produced highly vibrant chromatic accuracy, achieving a final SSIM of 0.9023, PSNR of 23.0104, and a highly competitive LPIPS score of 0.1769.
- **The Perception-Distortion Trade-off:** Our experiments conclusively demonstrated the Perception-Distortion Trade-off. In our pursuit of automated colorization, we successfully and deliberately traded 2% in pure structural similarity (stepping down from 0.928 in V1.4 to 0.902 in V1.6) to unlock profound improvements in human-perceived visual quality and vibrant chromatic accuracy.

7. Possible Improvements

While our current evaluation framework successfully relies on SSIM, PSNR, and LPIPS to balance structural integrity with human perception, a crucial avenue for future improvement is the integration of the Fréchet Inception Distance (FID) metric.

As demonstrated by Nathanael and Prasetyo (2024) in their evaluation of the Modified Multi-Attention U-Net, relying solely on pixel-level (PSNR/SSIM) or localized patch-level metrics can occasionally obscure macroscopic dataset-level qualities. FID solves this by extracting high-level feature representations using a pre-trained Inception-v3 network and calculating the Fréchet distance between the multivariate Gaussian distributions of the real and generated image features. The metric is mathematically defined as:

$$FID = ||\mu_r - \mu_g||^2 + Tr(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

where μ_r, μ_g denote the feature-wise means, and Σ_r, Σ_g denote the covariance matrices of the real and generated images, respectively.

Implementing FID will allow us to mathematically quantify the overall realism, diversity, and generative stability of our colorizations on a distribution level. Because FID heavily penalizes mode collapse, unnatural chromatic distributions, and GAN-induced artifacts, using it as a primary generative benchmark will further align our model's outputs with the high-fidelity realism achieved in contemporary state-of-the-art colorization research.

8. Conclusion

By systematically iterating through complex architectural bottlenecks, our team successfully developed an Attention-Enhanced U-Net capable of highly realistic automated image colorization. We documented the critical failures of utilizing Batch Normalization in color-generative tasks, solving the resulting color bleeding through the implementation of Group Normalization. Furthermore, we diagnosed and cured "Safe Color Collapse" by demonstrating that strict mathematical optimization (relying purely on standard L1 and PSNR) forces GANs to generate muddy, risk-averse sepia tones. By removing conflicting RGB-based perceptual losses,

handicapping the discriminator through TTUR, and utilizing Discriminator Feature Matching alongside the LPIPS metric, we successfully built a research-grade GAN that balances structural precision with vibrant, true-to-life human perception.

Appendix A. Detailed Roles

Fill in the table below (Table 1) for each teammate. Include the file names and number of lines of code written.

Appendix B. Code repository

GITHUB: <https://github.com/Cyube3/EC523Project>

References

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- 7) Miyato, T., Kataoka, T., Koyama, M., & Yoshida, Y. (2018). Spectral normalization for generative adversarial networks. *arXiv preprint arXiv:1802.05957*

Table 1. Team member contributions

Name	Task	File names	No. Lines of Code
Nicholas Chong	Implement dataset download and compression Implement V1.4 Implement V1.6 Implement V1.6 Ideate optimizing dataset retrieval	EC523ProjectColab.ipynb	48 435 410 439
Yeo Yong Yan	Implement V1.1 Implement V1.2 Implement V1.3	EC523ProjectColab.ipynb	423 462 432
Alexa	Implement V1.0	EC523ProjectColab.ipynb	344